



ELISA Validation Report

Determination of LT3114 in Rat K₂EDTA Plasma By ELISA

QPS Project Number: 344-1401
Client Project Number: NA

Sponsor

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ELISA VALIDATION REPORT

REPORT TITLE:

Determination of LT3114 in Rat K₂EDTA Plasma by ELISA

QPS PROJECT NUMBER:

344-1401

VALIDATION INITIATED:

9 Jul 2014

CONTRIBUTING SCIENTISTS:

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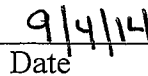
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1. SIGNATURE PAGE

WRITTEN BY:

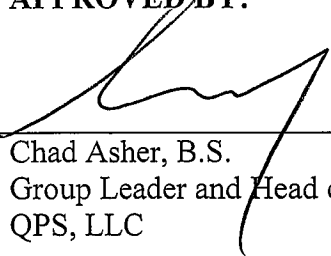


Melissa Calvarese, B.S.
Principal Investigator
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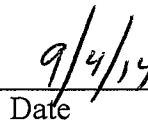


Date

APPROVED BY:



Chad Asher, B.S.
Group Leader and Head of PK Bioanalysis
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Date

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2. COMPLIANCE STATEMENT

This portion of the study was conducted in compliance with the Food and Drug Administration (FDA) Good Laboratory Practice Regulations (GLP) as set forth in Title 21 of the U.S. Code of Federal Regulations Part 58; the Organization for Economic Co-operation and Development (OECD) Principles of Good Laboratory Practice [C (97) 186/Final]; the Japanese Ministry of Health, Labor, and Welfare (MHLW); Ordinance No. 21, March 26, 1997; and the FDA Guidance for Industry: Bioanalytical Method Validation, Sept 2013.

To the best of my knowledge this report accurately reflects the raw data. I have reviewed the study and agree that the data supports the conclusions stated herein.

Melissa Calvarese

Melissa Calvarese, B.S.

Principal Investigator

QPS, LLC

9/4/14

Date

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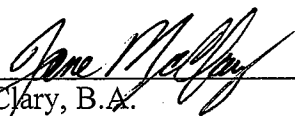
3. QUALITY ASSURANCE STATEMENT

QPS 344-1401

Protocol Title: Determination of LT3114 in Rat K₂EDTA Plasma by ELISA

This QPS study has been audited and found to accurately reflect the requirements set forth by the 344-1401 validation protocol and standard operating procedures, and the results reported accurately reflect the source data. The following table summarizes the inspections and audits performed throughout the course of the study and the dates when each audit report and any findings were reported by QPS Quality Assurance.

Audit Description	QPS Auditor	Audit Dates	Date Audit Findings Reported to QPS Principal Investigator and Management
Draft Protocol	S. Ma	02 Jul 2014	02 Jul 2014
Study Records/ Draft Report	J. McClary	11,12 Aug 2014	12 Aug 2014



Jane McClary, B.A.
Quality Assurance
QPS, LLC



Date

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4. VALIDATION SUMMARY FOR LT3114

An immunoassay method for the determination of LT3114 in rat K₂EDTA plasma was developed and validated. Experiments were conducted at QPS to evaluate the precision and accuracy of the assay in rat plasma.

The ELISA assay utilized a sandwich format, in which, microtiter plates are pre-coated with LPA-BSA Conjugate. The plate is washed. Samples (including standards and QCs) are added and incubated on the plate. LT3114 is detected by HRP conjugated goat anti-human IgG monkey adsorbed. TMB is used as a substrate for the HRP. The reaction is stopped with 1N HCl.

Assay range, intra-batch and inter-batch precision / accuracy data, along with the stability, dilution linearity, and matrix evaluation data are provided in the following summary table. Additional long term frozen stability will be evaluated at a later date, and will be added as an addendum.

Report Title	Determination of LT3114 in Rat K ₂ EDTA Plasma by ELISA
Report Number	QPS 344-1401
Analyte Name	LT3114
Sample Volume:	5µL (Before 100x dilution, duplicate measurement)
Analytical Method Type	ELISA
Calibration Range:	100 – 2000 ng/mL
QC Concentrations	100, 300, 750, 1500, and 2000 ng/mL
QC Intra-batch Precision (%CV)	4.1% - 7.1%
QC Intra-batch Accuracy (%RE)	1.1% - 18.0%
QC Inter-batch Precision (%CV)	4.8% - 8.5%
QC Inter-batch Accuracy (%RE)	-0.7% - 11.3%
Room Temperature Stability in Rat K ₂ EDTA Plasma	24 Hours at Room Temperature
Freeze/thaw Stability in Rat K ₂ EDTA Plasma	5 Cycles at -70°C and -20°C/ Room Temperature
Long-term Storage Stability in Rat K ₂ EDTA Plasma	30 Days at -20°C and -70°C
Dilution Linearity	900,000 ng/mL diluted 1, 100, 500, 1000, and 5000 -fold
Selectivity (10 lots, spiked 100, 2000 ng/mL)	100% lots tested within 100±25% Recovery

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5. OBJECTIVE

The purpose of this study was to validate an ELISA to determine the concentrations of LT3114 in rat K₂EDTA plasma samples collected during a nonclinical study.

6. ABBREVIATIONS AND DEFINITIONS

This section provides abbreviations and definitions of terms and concepts that may be commonly used throughout this report.

AQL	Above Quantifiable Limit
BQL	Below Quantifiable Limit
Exc	Sample excluded according to QPS Validation Protocol 344-1401
LLOQ	Lower Limit of Quantification
LLOQ QC	Lower Limit of Quantification Quality Control
MRD	Minimum required Dilution
NC	Not Calculated
NA	Not Applicable
NR	Not Reported
OD	Optical Density
QC	Quality Control
r²	Coefficient of Determination
SD	Standard Deviation
SOP	Standard Operating Procedure
ULOQ	Upper Limit of Quantification
ULOQ QC	Upper Limit of Quantification Quality Control

$$\text{Precision} \quad \% \text{Coefficient of Variation (\%CV)} = 100 \times \frac{\text{Standard Deviation}}{\text{Mean Concentration}}$$

$$\text{Accuracy} \quad (\% \text{RE}) = 100 \times \frac{\text{Mean Found Concentration} - \text{Nominal Concentration}}{\text{Nominal Concentration}}$$

Calibration standard and Quality control samples at five concentration levels were prepared in matrix and were determined based upon the actual concentrations of the stock solutions.

All statistics in the data tables are calculated by Watson™ LIMS or based on the “precision as displayed” option of Microsoft® Excel (e.g., Mean, S.D., %CV, %Diff).

7. ANALYTE INFORMATION

Name: LT3114
Concentration: 20.14^a mg/mL
Purity: 96%
Supplied Form Lpath
Lot Number: LP121416
Expiration Date: 4/30/16 (assigned per QPS SOP LP-007)^b
Storage Conditions: -70°C

^aWith purity applied, concentration will be used at 19.33mg/mL.

^bNo supporting documentation available for stability or retest date of compound at current storage condition.

8. ACCEPTANCE CRITERIA

The validation acceptance criteria and the statistical data will be determined at a minimum to QPS' SOPs TLM-001 Validation of Binding Assays and QPS validation protocol 344-1401. Additional experiments (if any) not governed under QPS' SOPs will follow the instructions provided by the sponsor.

Six successful, complete precision and accuracy runs (out of which three are consecutive runs) are required to validate a method. The evaluation of an intra-batch validation run includes a minimum of six replicates of QCs at each concentration and a minimum of five QC concentrations (including LLOQ and ULOQ). The evaluation of inter-batch validation runs (minimum of six runs) includes a minimum of three replicates of QCs at each concentration per run. Each run must include a minimum of five QC concentrations.

Method Acceptance: A method is accepted if it successfully fulfills the following:

- Intra-batch Accuracy and Precision acceptance criteria: For each intra-batch validation run, the mean of six or more replicates at each concentration of the QC samples is not allowed to deviate more than $\pm 20\%$ ($\pm 25\%$ at LLOQ and ULOQ QCs) from its nominal value. In addition, the intra-batch precision at each concentration must not exceed 20% (25% at LLOQ and ULOQ QCs) when calculated as the %CV.
- Inter-batch Accuracy and Precision acceptance criteria: The mean value at each QC concentration must not exceed $\pm 20.0\%$ ($\pm 25.0\%$ at LLOQ and ULOQ) from its nominal value. In addition, the inter-batch precision at each concentration must not exceed 20% (25% at LLOQ and ULOQ QCs) when calculated as the %CV.
- Dilution test acceptance criteria: The mean of the back calculated concentration (after the dilution factor has been applied) must be within

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100±20% of the nominal value, and the %CV of the calculated concentrations must not exceed 20% to be considered acceptable.

- Assay selectivity acceptance criteria: At least 80% of the lots spiked with LT3114 should have recovery between 75-125%.
- Room temperature and freeze/thaw stability acceptance criteria: The mean back calculated concentration must be within 100±20% of the nominal value, and the %CV of the calculated concentrations must not exceed 20% to be considered stable.
- Frozen stability samples acceptance criteria: The back calculated concentration is compared to the nominal value of the QC samples, which was confirmed in the initial analysis as properly prepared. The mean value for each concentration level must be within 100±20% of the nominal value, and the %CV of the calculated concentrations must not exceed 20% to be considered stable.

Run Acceptance criteria:

- For QCs, validation samples and future study samples, each replicate will be measured twice and the average will be reported if the %CV from the two measurements is $\leq 25\%$. Otherwise, the value from the two measurements will not be reported.
- Production runs or sample analysis runs are accepted based on the performance of calibration standards and QCs. Only standards back-calculated to over $\pm 20\%$ ($\pm 25\%$ at LLOQ and ULOQ) different from their nominal concentrations will be excluded from the regression analysis. If more than one standard back calculates to over $\pm 20\%$ ($\pm 25\%$ at LLOQ and ULOQ), exclusion should be carried out one standard at a time to see if the other standards would fall within $\pm 20\%$ ($\pm 25\%$ at LLOQ and ULOQ) of the nominal value after the exclusion.

If both of the calibration standards at the lowest (or highest) concentration of the curve are excluded from the regression analysis due to various reasons (e.g., not enough sensitivity at the lower end, curve bend at the higher end, QCs or standards at the specific concentration fail to meet acceptance criteria), the LLOQ (or ULOQ) shall be moved up (or down) to the next standard concentration in sequence (also known as the truncation of the standard curve).

Finally, a minimum of six standard concentrations must be within the assay range. Excluded calibration standards that fall outside of the calibration range shall not be used to evaluate the acceptability of the run. All excluded

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calibration standards over $\pm 20\%$ ($\pm 25\%$ at LLOQ and ULOQ) within the calibration range should be reported (e.g., as Exc) and should be included when evaluating the acceptability of the run.

Anchor points may be included or excluded to facilitate curve fitting based on scientific judgment.

At least 2/3 of the back calculated QC concentrations must be within $100 \pm 20\%$ ($100 \pm 25\%$ at the LLOQ and ULOQ QCs) of their nominal value, 50% of QC at each concentration must not deviate more than $\pm 20\%$ ($\pm 25\%$ at LLOQ and ULOQ QCs) from its nominal value.

- Special case of run acceptance for accuracy and precisions run:
The precision and accuracy runs are accepted solely based on the performance of the calibration curve. A minimum of 75% of the total number of calibration standards in the calibration range (between the LLOQ and ULOQ of the run) should be within $100 \pm 20\%$ ($100 \pm 25\%$ at LLOQ and ULOQ standards) of their nominal value, and a minimum of six unique standard concentrations must be within the assay range. The calibration curve must contain at least one calibration standard at both the LLOQ and ULOQ of the range.

9. MATERIALS

9.1. Sample-Handling Equipment

- 96-well microtiter plates (Costar Cat #2592)
- 96-well format micro titer/dilution tubes
- 25-100 mL Reagent reservoir
- Plate sealers
- Polypropylene plastic tubes: various volumes
- Pipette tips for single and multichannel pipettes
- Graduated cylinder: various volumes
- 8-Channel StatMatic II microplate dispenser

9.2. Analytical Equipment

Spectramax Microplate Reader (Molecular Devices).

10. METHODS

10.1. Preparation of Secondary Stock Solutions:

Two sets of stock solutions were prepared separately for standards and QCs. The prepared stock solutions were used immediately to prepare subsequent working standard and QC solutions and the remaining portion was discarded.

10.2. Preparation of Working Standard Solutions in Rat K₂EDTA Plasma

Appropriate amount of stock was combined with rat plasma according to the following table to make the calibration standard working solutions. Each concentration of the standards

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contains at least 95% (v/v) rat plasma. The prepared working standard solutions were aliquoted for single use and were stored at -70°C. The blank pooled rat plasma served as the zero standard (Blank).

Concentration (ng/mL)	Vol. of Stock 4000 ng/mL (μL)	Vol. of RTPL (μL)	Total Volume (μL)
0	0	200	200
50	5	395	400
100	5	195	200
200	10	190	200
400	20	180	200
800	40	160	200
1000	50	150	200
1400	70	130	200
1800	90	110	200
2000	100	100	200
4000	200	0	200

10.3. Preparation of Working QC Solutions in Rat K₂EDTA Plasma

Appropriate amount of stock was combined with rat plasma according to the following table to make the QC working solutions. Each concentration of the QCs contains at least 95% (v/v) rat plasma. The prepared QC solutions were aliquoted for single use and were stored at -70°C.

Concentration (ng/mL)	Vol. of Stock 4000 ng/mL (μL)	Vol. of RTPL (μL)	Total Volume (μL)
100	5.0	195.0	200
300	75.0	925.0	1000
750	37.5	162.5	200
1500	375	625.0	1000
2000	100	100.0	200

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10.4. Analytical Method

<u>Parameter</u>	<u>Value of Parameter</u>
General:	
Analyte	LT3114
Matrix	Rat K ₂ EDTA Plasma
Calibration standard concentrations	50(AP), 100, 200, 400, 800, 1000, 1400, 1800, 2000 and 4000(AP) ng/mL
Quality Control Concentrations	100, 300, 750, 1500, and 2000 ng/mL
Minimum required dilution	100x in Dilipidized Bovine Serum
Sample volume	5 µL (before dilution, 2 wells) 100 µL/well (after dilution)
ELISA procedure summary	Microtiter plates are pre-coated with LPA-BSA. After washing, samples (including standards and QCs) are added and incubated on the plate. LT3114 is detected by horseradish peroxidase conjugated goat anti-human IgG H and L, Monkey Adsorbed Ab. TMB is used as a substrate for the HRP. The reaction is stopped with 1N HCl. The color intensity is proportional to the quantity of LT3114.
Data Acquisition:	
Equipment	Molecular Devices SpectramaxPLUS ³⁸⁴
Program	Softmax Pro Version 3.1.2
Reading mode	Endpoint
Wavelength	450 nm (reading), 630 nm (reference)
Calibration	Auto-calibration on
Blanking	No pre-read, Plate blank not used
Mixing	Off
Analysis and Reporting:	
Program	Watson™ LIMS Version 7.4.1
Regression type	4-parameter (auto-estimate)
Weighting:	1
Anchoring Point	Use as needed.
Precision for Reporting:	3 significant digits

11. RESULTS AND DISCUSSION

Table 1 lists all the validation runs performed.

11.1. Regression Analysis

Calculations were done using 4-parameter (auto-estimate) regression using Watson™ LIMS (v. 7.4.1, Thermo Electron Corporation, USA) Table 2 lists the back-calculated calibration standard concentrations for LT3114 and Table 3 lists the standard curve parameters for

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LT3114. The results show a 4-Parameter fit from 100-2000 ng/mL for LT3114. Figure 1 shows a typical calibration curve of LT3114 in rat plasma.

11.2. Precision and Accuracy

Precision and accuracy of the method were determined by analyzing QC samples at a minimum of five different concentrations within the standard curve range.

Precision was determined by the coefficient of variation (%CV) and accuracy was determined by percent recovery (%RE). The acceptable maximum %CV for a valid assay is 25% at all concentrations. The acceptable accuracy (%RE) for a valid assay is within $100 \pm 20\%$ (25% at LLOQ and ULOQ QCs) of the nominal concentration.

The intra-batch precision and accuracy within an analytical run were determined by analyzing six replicates for each QC sample including the ULOQ and the LLOQ QC from the same run. The results for intra-batch precision and accuracy (Table 4) were acceptable.

The inter-batch precision and accuracy of the assay over different analytical runs were determined by analyzing QC samples from six validation runs with three replicates in each run. The results for inter-batch precision and accuracy (Table 5A) were acceptable. In addition the results for QCs from all runs are summarized in Table 5B.

11.3. Lower Limit of Quantification

The lower limit of quantification (LLOQ) of the assay was 100 ng/mL for LT3114. For sample analysis, concentrations found below the LLOQ after the minimum required dilution will be reported as 'BQL', if applicable, since the precision and accuracy of the assay were determined to that limit.

11.4. Stability

11.4.1. Room Temperature Stability

Room temperature stability of LT3114 in rat plasma was performed to evaluate LT3114 stability in rat plasma for the time that the samples would be at room temperature during sample handling and processing. Stability of the QC samples was determined at two concentrations (low and high QCs) in minimum three replicates at both 6 and 24 hours. The results (Table 6) indicate that LT3114 in rat plasma is stable at room temperature for at least 24 hours.

11.4.2. Freeze/Thaw Stability

The stability of LT3114 in rat plasma after five freeze/thaw cycles was determined. QC samples at two concentrations (Low and high QCs) were frozen at -20°C or -70°C (for a minimum of 24 hours for cycle 1 and a minimum of 12 hours for later cycles) and thawed at room temperature. This freeze/thaw cycle was repeated five times. After the completion of the fifth freeze/thaw cycle, the samples were analyzed and compared to the nominal value. The results (Table 7) indicate LT3114 is stable for five freeze/thaw cycles in rat plasma under the described conditions.

11.5. Long-Term Frozen Storage Stability

LT3114 quality control samples in rat plasma were stored at -70°C or -20°C for 30 days before being analyzed against a freshly prepared calibration standard curve. The results (Table 8) indicate LT3114 is stable for at least 30 days in rat plasma at -70°C and -20°C. The frozen storage stability of LT3114 in rat plasma at -70°C and -20°C at longer periods will be assessed at a later date and an addendum to the report will be provided.

11.6. Dilution Linearity and Hook Effect

The purpose of the dilution test was to ensure that samples could be diluted with rat plasma without affecting the final calculated concentration. Rat plasma samples prepared at one concentration 900,000 ng/mL were diluted with rat plasma at dilution factors of 1, 100, 500, 1000, and 5000 in three replicates at each level. The results were corrected with the dilution factor and compared to the nominal concentration. The results in Table 9 show that the difference between the mean of the adjusted concentration (found concentration multiplied by dilution factor) and the nominal concentration of LT3114 was within the acceptable range. There was no hook effect observed.

11.7. Assay selectivity in Rat K₂EDTA Plasma

Ten individual lots of rat plasma were spiked with or without LT3114 and measured for LT3114 concentration. Table 10 indicates that 100% of the tested lots had a recovery between 75-125%, which is acceptable.

12. STUDY SPECIFIC DEVIATIONS OR OBSERVATIONS

DEV-14-376: In step 8 “Plate Reading” the Analytical Procedure specified Lm2 as 570nm, however the plates were all read using 630nm as Lm 2. The method development for this project was consistent with the 630nm wavelength used during validation. Throughout the study analyst and PI incorrectly initial dated that the 570nm was used as the Lm2. There is no impact on study data and a note to the record was added the study notebook for clarification.

13. CONCLUSION

An ELISA method (QPS method 344-1401) has been validated for the quantitation of LT3114 in rat K₂EDTA plasma from 100 – 2000 ng/mL. With a validated dilution factor of 5000, this assay is suitable for measuring LT3114 in rat K₂EDTA plasma from 100 – 10000000 ng/mL.

14. REFERENCES

Method Validation Protocol:

QPS Project No. 344-1401

SOPs:

QPS LP-004: “Analytical Standards and Quality Controls”

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QPS LP-007: “Analytical Reference Compounds”

QPS LP-034: “Receipt and Handling of Biological Matrices”

QPS TLM-001: “Validation of Quantitative Ligand Binding Assays”

Notebooks:

QPS validation notebook 344-1401

LT3114 assay notebook

15. ARCHIVE STATEMENT

All raw data (including electronic SoftmaxPro files), a copy of this bioanalytical report and required supporting information for this study will be held under the control of QPS, LLC, 3 Innovation Way, Suite 240, Newark DE, 19711, USA. Study records will be archived according to the most current version of SOP GN-014, “Archives”

16. TABLES

Table 1 Validation Run Log

Curve Number	Analysis Date	Run Description	Accepted/ Rejected ^a
1	9-Jul-14	140709-3441401-RTPL-EEQ-1A-001	Accepted
2	9-Jul-14	140709-3441401-RTPL-EEQ-2B-002	Accepted
3	10-Jul-14	140710-3441401-RTPL-GP-1A-003	Accepted
4	10-Jul-14	140710-3441401-RTPL-GP-2B-004	Accepted
5	10-Jul-14	140710-3441401-RTPL-MAC-1A-005	Accepted
6	10-Jul-14	140710-3441401-RTPL-MAC-2B-006	Accepted
7	11-Jul-14	140711-3441401-RTPL-EEQ-1A-007	Accepted
8	11-Jul-14	140711-3441401-RTPL-EEQ-2B-008	Accepted
9	15-Jul-14	140715-3441401-RTPL-MAC-1A-009	Accepted
10	7-Aug-14	140807-3441401-RTPL-MAC-1A-010	Accepted

a: Acceptance based on QPS Validation Protocol 344-1401

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**Table 2 Back-Calculated Calibration Standard Concentrations of LT3114 Rat
K₂EDTA Plasma**

Analysis Date	Curve Number	Concentrations (ng/mL)							
		100	200	400	800	1000	1400	1800	2000
9-Jul-14	1	98.5	200	422	805	1020	1390	1830	2000
		97.5	203	420	786	974	1360	1810	2010
9-Jul-14	2	102	195	407	800	989	1440	1820	2010
		101	195	411	788	977	1400	1800	1960
10-Jul-14	3	97.0	205	393	760	1020	1440	1790	2050
		103	208	423	773	990	1450	1780	1940
10-Jul-14	4	98.4	195	411	812	1010	1380	1780	2040
		Exc	200	403	806	1000	1370	1840	1960
10-Jul-14	5	107	211	435	814	998	1380	1920	2010
		99.0	204	429	786	950	1370	1780	1960
10-Jul-14	6	107	199	418	799	981	1410	1860	1980
		100	203	403	809	982	1370	1810	1980
11-Jul-14	7	98.4	200	403	812	992	1420	1780	2050
		99.7	193	403	803	990	1390	1800	1980
11-Jul-14	8	101	203	408	789	963	1400	1840	1930
		102	204	412	803	998	1410	1830	2030
15-Jul-14	9	96.8	193	387	799	1010	1400	1790	1980
		116	197	395	793	1010	1430	1800	2000
7-Aug-14	10	104	196	402	791	980	1380	1780	1990
		104	202	409	807	1010	1430	1770	2050
n		19	20	20	20	20	20	20	20
Mean		102	200	410	797	992	1400	1810	2000
S.D.		4.62	4.97	12.1	13.8	18.5	26.9	35.9	36.2
%CV		4.5	2.5	3.0	1.7	1.9	1.9	2.0	1.8
%RE		2.0	0.0	2.5	-0.4	-0.8	0.0	0.6	0.0

Exc: Excluded according to QPS Validation Protocol 344-1401

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Table 3 LT3114 Standard Curve Regression Parameters

Analysis Date	Curve Number	Min	Max	Slope	Ed50	R-Squared	LLOQ	ULOQ	Regression Footnote(s)
9-Jul-14	1	0.079423676	9.659042685	1.108156444	12960.072455515	0.9998	100	2000	1
9-Jul-14	2	0.055879541	12.406488020	1.060574786	18980.594965749	0.9998	100	2000	1
10-Jul-14	3	0.063566904	10.727660359	1.109855676	16126.977200774	0.9995	100	2000	1
10-Jul-14	4	0.069832368	1167.099518788	1.033531956	1890030.078981170	0.9990	100	2000	1
10-Jul-14	5	0.081810865	6.740268788	1.178579073	8952.648116197	0.9990	100	2000	1
10-Jul-14	6	0.088745396	10.678073164	1.113933308	13960.505017757	0.9997	100	2000	1
11-Jul-14	7	0.067853005	7.573096458	1.072305456	7448.187213060	0.9998	100	2000	1
11-Jul-14	8	0.086653370	6.527260072	1.143864548	5475.132820635	0.9994	100	2000	1
15-Jul-14	9	0.061798590	17.833798412	1.025876493	24022.727523934	0.9993	100	2000	1
7-Aug-14	10	0.112457337	10.473655501	1.053428960	12282.079336779	0.9997	100	2000	1

Regression Footnote(s):

1) Resp. = (Min - Max) / (1 + (Conc. / Ed50) ** Slope) + Max

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Table 4 Intra-Batch Precision and Accuracy of LT3114 in Rat K₂EDTA Plasma

Analysis Date	Curve Number	Concentrations (ng/mL)				
		100	300	750	1500	2000
9-Jul-14	1	129	374	817	1420	2240
		112	343	790	1470	2160
		110	339	739	1500	2030
		114	342	719	1540	1980
		114	356	733	1590	1980
		109	367	748	1730	2010
n		6	6	6	6	6
Intrarun Mean		115	354	758	1540	2070
Intrarun SD		7.31	14.6	37.7	109	108
Intrarun %CV		6.4	4.1	5.0	7.1	5.2
Intrarun %RE		15.0	18.0	1.1	2.7	3.5

Table 5A Inter-Batch Precision and Accuracy of LT3114 in Rat K₂EDTA Plasma

Analysis Date	Curve Number	Concentrations (ng/mL)				
		100	300	750	1500	2000
9-Jul-14	1	129	374	817	1420	2240
		112	343	790	1470	2160
		110	339	739	1500	2030
		114	342	719	1540	1980
		114	356	733	1590	1980
		109	367	748	1730	2010
9-Jul-14	2	103	329	772	1500	2050
		98.0	291	701	1400	1880
		108	305	754	1500	2040
10-Jul-14	3	103	318	816	1470	1990
		112	319	722	1370	1930
		115	348	857	1610	2010
10-Jul-14	4	188 ^a	378	867	1530	2090
		95.4	330	775	1480	2140
		108	317	793	1560	2040
10-Jul-14	5	101	338	807	1530	2120
		92.5	316	714	1370	1900
		107	325	813	1430	1930
10-Jul-14	6	99.2	319	785	1480	1990
		93.3	328	735	1390	1840
		NR	328	829	1510	2020
n		19	21	21	21	21
Mean		106	334	776	1490	2020
S.D.		8.97	22.0	47.8	86.4	96.6
%CV		8.5	6.6	6.2	5.8	4.8
%RE		6.0	11.3	3.5	-0.7	1.0

NR: Not Reportable %CV>25%

a: Value excluded from calculations (Value > 3x S.D.)

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Table 5B **LT3114 Quality Controls in Rat K₂EDTA Plasma for All Runs**

Analysis Date	Curve Number	Concentrations (ng/mL)				
		100	300	750	1500	2000
9-Jul-14	1	129	374	817	1420	2240
		112	343	790	1470	2160
		110	339	739	1500	2030
		114	342	719	1540	1980
		114	356	733	1590	1980
		109	367	748	1730	2010
9-Jul-14	2	103	329	772	1500	2050
		98.0	291	701	1400	1880
		108	305	754	1500	2040
10-Jul-14	3	103	318	816	1470	1990
		112	319	722	1370	1930
		115	348	857	1610	2010
10-Jul-14	4	188 ^a	378	867	1530	2090
		95.4	330	775	1480	2140
		108	317	793	1560	2040
10-Jul-14	5	101	338	807	1530	2120
		92.5	316	714	1370	1900
		107	325	813	1430	1930
10-Jul-14	6	99.2	319	785	1480	1990
		93.3	328	735	1390	1840
		NR	328	829	1510	2020
11-Jul-14	7		308	731	1450	
			302	704	1530	
11-Jul-14	8		320	760	1440	
			305	736	1490	
15-Jul-14	9		339	750	1510	
			287	740	1570	
7-Aug-14	10		303	755	1490	
			325	838	1660	
n		19	29	29	29	21
Mean		106	328	769	1500	2020
S.D.		8.97	22.8	46.1	81.9	96.6
%CV		8.5	7.0	6.0	5.5	4.8
%RE		6.0	9.3	2.5	0.0	1.0

NR: Not Reportable %CV>25%

a: Value excluded from calculations (Value > 3x S.D.)

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Table 6 Room Temperature Stability of LT3114 in Rat K₂EDTA Plasma

	Nominal Concentration (ng/mL)	
	300 Lot: MAC140708-12	1500 Lot: MAC140708-14
6 hours	NR 284 289	1410 1380 1450
n	2	3
Mean	287	1410
SD	3.54	35.1
%CV	1.2	2.5
%RE	-4.3	-6.0
24 hour	260 250 256	1320 1260 1330
n	3	3
Mean	255	1300
SD	5.03	37.9
%CV	2.0	2.9
%RE	-15.0	-13.3

NR: Not Reportable %CV>25%

Run ID: 140715-3441401-RTPL-MAC-1A-009

Table 7 Freeze/Thaw Stability of LT3114 in Rat K₂EDTA Plasma

	Nominal Concentration (ng/mL)	
	300 Lot: MAC140708-12	1500 Lot: MAC140708-14
-70°C	274	1370
5 Cycles	274	1400
	273	1470
N	3	3
Mean	274	1410
SD	0.577	51.3
%CV	0.2	3.6
%RE	-8.7	-6.0
-20°C	287	1350
5 Cycles	277	1360
	272	1360
n	3	3
Mean	279	1360
SD	7.64	5.77
%CV	2.7	0.4
%RE	-7.0	-9.3

Run ID: 140715-3441401-RTPL-MAC-1A-009

Table 8 30 Day Long Term Frozen Stability of LT3114 in Rat K₂EDTA Plasma

	Nominal Concentration (ng/mL)	
	300 Lot: MAC140708-12	1500 Lot: MAC140708-14
-20°C	304	1580
	304	1540
	319	1590
N	3	3
Mean	309	1570
SD	8.66	26.5
%CV	2.8	1.7
%RE	3.0	4.7
-70°C	323	1470
	323	1570
	321	1570
n	3	3
Mean	322	1540
SD	1.15	57.7
%CV	0.4	3.7
%RE	7.3	2.7

Run ID: 140807-3441401-RTPL-MAC-1A-010

Table 9 Dilution Test of LT3114 in Rat K₂EDTA Plasma

Nominal Concentration (ng/mL)	Dilution Factor	LT3114 (ng/mL)	
900000	1	AQL	
		AQL	
		AQL	
		n	3
		Mean	NC
		SD	NC
	100	%CV	NC
		%RE	NC
		AQL	AQL
		AQL	AQL
		n	3
		Mean	NC
	500	SD	NC
		%CV	NC
		%RE	NC
		994000	994000
		979000	979000
		n	3
	1000	Mean	989000
		SD	8660
		%CV	0.9
		%RE	9.9
		1090000	975000
		975000	1030000
	5000	1030000	3
		Mean	1030000
		SD	57500
		%CV	5.6
		%RE	14.4
		975000	942000
		942000	959000
		959000	3
		n	3
		Mean	959000
		SD	16500
		%CV	1.7
		%RE	6.6
		Cumulative n	9
		Cumulative Mean	993000
		Cumulative SD	43800
Cumulative %CV		4.4	
Cumulative %RE		10.3	

Run ID: 140711-3441401-RTPL-EEQ-1A-007

AQL: Above Limit of Quantitation (>2000 ng/mL)

NC: Not calculated

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Table 10 Assay Selectivity of LT3114 in Rat K₂EDTA Plasma

Individual Rat Plasma	LT3114 Concentration (ng/mL)				
	Blank (0 ng/mL)	Spike-in LLOQ QC (100 ng/mL)		Spike-in ULOQ QC (2000 ng/mL)	
Lot #	Found	Found	% Recovery of Spiked Lot	Found	% Recovery of Spiked Lot
RAT236133	BQL	106	106.0	1980	99.0
RAT236134	BQL	109	109.0	2060	103.0
RAT236135	BQL	107	107.0	2100	105.0
RAT236136	BQL	101	101.0	2050	102.5
RAT236137	BQL	100	100.0	2020	101.0
RAT236138	BQL	106	106.0	1980	99.0
RAT236139	BQL	105	105.0	2020	101.0
RAT236140	BQL	106	106.0	2010	100.5
RAT236141	BQL	103	103.0	2020	101.0
RAT236142	BQL	103	103.0	2010	100.5
n		10		10	
Mean		105		2030	
SD		2.80		36.6	
%CV		2.7		1.8	
%RE		5.0		1.5	

Run ID: 140711-3441401-RTPL-EEQ-2B-008

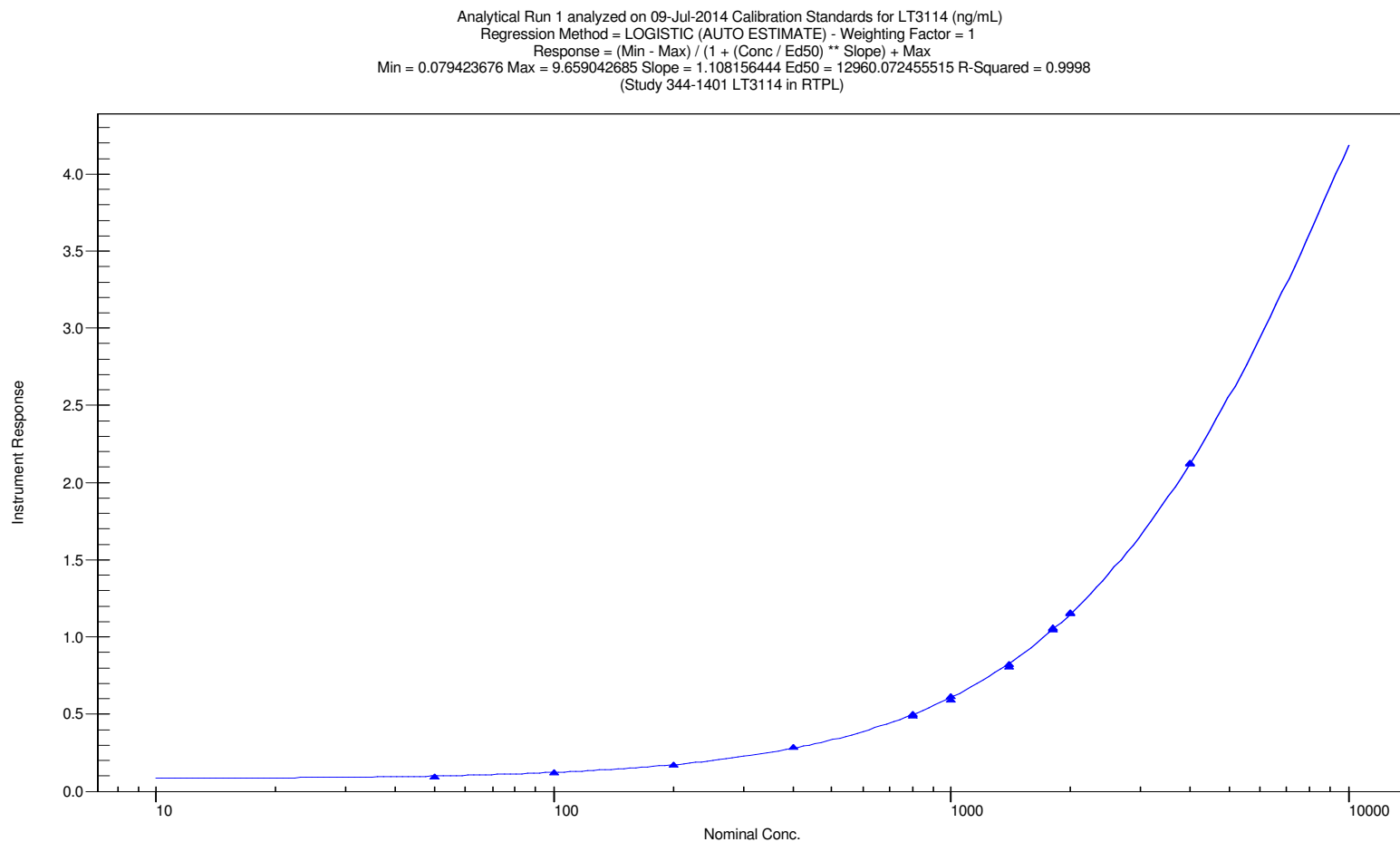
BQL: Below Limit of Quantitation (<100 ng/mL)

% Recovery: (Found Concentration / Nominal of the Spiked) X 100%

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17. FIGURES

Figure 1 **Typical Standard Curve for LT3114 ELISA in Rat K₂EDTA Plasma**



Run ID: 140709-3441401-RTPL-EEQ-1A-001

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30

18. APPENDICES

Appendix A Certificate of Analysis of LT3114

		Certificate of Analysis Form			
TITLE	Certificate of Analysis for Monoclonal Antibody Produced by Lpath				LOT NUMBER
					LP121416
DOCUMENT NUMBER	REVISION	SUPERCEDED REVISION	ISSUE DATE	EFFECTIVE DATE	PAGE
FM-COA-13	0.0.1	0.0.0	OCT 31, 2013	Nov 1, 2013	1 of 5

Name	Part Number	Lot Number
Humanized Lpathomab	LT3114	LP121416
Formulation Buffer		Storage Temperature
PBS (Mediatech #21-040, Lot# 21040283)		2 to 8 °C
Production Date	Retest Date	Project /Use
5-10-2013	N/A	TBD

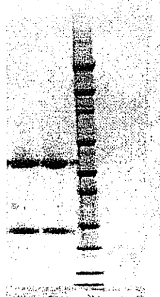
Summary
<u>Production location, date, and method</u> The antibody was produced from stable CHO clone (191+192) 3114 3B8/1B4/2F1 in 4x 3-L shake flasks at Lpath. Takedown on d10, May 10, 2013. Pooled, clarified harvest was stored at -20 °C. Lot LP121416 consists of 3x 1-L bottles of supe which cracked during storage. Lot LP121417 consists of 1x 1-L bottle of supe which did not crack during storage. Purified lots LP121416 and LP121417 were pooled to create lot LP121419. <u>Purification location, date and method</u> Antibody purification was initiated on August 21, 2013 at Lpath using affinity and AEX chromatography (MabSelect Protein A and Q membrane Sartorius #92IEXQ42D4-OOA, respectively). Following concentration, the antibody was dialyzed against 3x 2-L PBS (Mediatech) and passed through a 0.22 um filter flask unit (Corning). <u>Concentration by A₂₈₀ and extinction coefficient</u> The final sample is 21.5 mg/mL as determined by UV spectroscopy using a theoretical extinction coefficient of 1.4 mL/mg. <u>Total amount of Ab (mg)</u> 1400 mg <u>Appearance details:</u> Clear, slightly yellow solution

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Pages 1-5
Rcm 5/2/14

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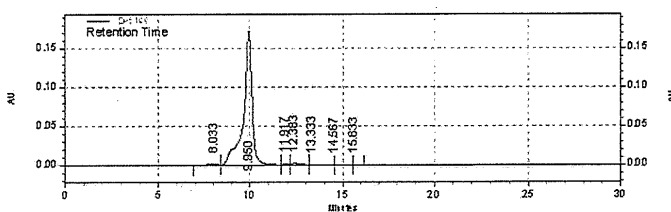
		Certificate of Analysis Form			
TITLE	Certificate of Analysis for Monoclonal Antibody Produced by Lpath				LOT NUMBER
					LP121416
DOCUMENT NUMBER	REVISION	SUPERCEDED REVISION	ISSUE DATE	EFFECTIVE DATE	PAGE
FM-COA-13	0.0.1	0.0.0	OCT 31, 2013	Nov 1, 2013	2 of 5

Attribute	Test	Specifications	Results
Identity	Appearance	Particulate free, clear solution Colorless or slightly yellow	Particulate free, clear solution Slightly yellow
	SDS-PAGE (Non-Reduced)	Report Results	Major band at 165 kDa
	SDS-PAGE (Reduced)	Two pronounced bands; One at ~50 kDa; One at ~ 25 kDa	Two pronounced bands; One at ~55 kDa; One at ~ 28 kDa
	Insert images of SDS-PAGE	Non-Reduced LT3114 new 2.7 µg LT3114 new 2.7 µgR Standard Empty	Reduced LT3114 new 2.7 µg LT3114 new 2.7 µg Standard Empty
			
	Isoelectric Focusing (by IEF Gel)	Report Results	Not done

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		Certificate of Analysis Form			
TITLE	Certificate of Analysis for Monoclonal Antibody Produced by Lpath				LOT NUMBER
					LP121416
DOCUMENT NUMBER	REVISION	SUPERCEDED REVISION	ISSUE DATE	EFFECTIVE DATE	PAGE
FM-COA-13	0.0.1	0.0.0	OCT 31, 2013	Nov 1, 2013	3 of 5

Attribute	Test	Specifications	Results																																																			
Potency	Protein Concentration (OD-280)	Report Results	20.14 mg/mL																																																			
	LPA Binding Direct ELISA (EC 50)	Report Results	25.05 ng/mL																																																			
	Bioactivity Assay (IL-6 and EDG2)	Positive Result	Does not stimulate IL-6 EDG2 IC50 = 13.65 µg/mL																																																			
Purity	Insert image of SE-HPLC																																																					
		<table><tr><th colspan="5">Det166 Results</th></tr><tr><th>Time</th><th>Area</th><th>Area %</th><th>Height</th><th>Height %</th></tr><tr><td>8.033</td><td>68097</td><td>1.14</td><td>1413</td><td>0.78</td></tr><tr><td>9.950</td><td>5707101</td><td>95.95</td><td>173377</td><td>96.24</td></tr><tr><td>11.917</td><td>35431</td><td>0.60</td><td>1744</td><td>0.97</td></tr><tr><td>12.383</td><td>97022</td><td>1.63</td><td>2645</td><td>1.47</td></tr><tr><td>13.333</td><td>31081</td><td>0.52</td><td>668</td><td>0.37</td></tr><tr><td>14.567</td><td>8116</td><td>0.14</td><td>234</td><td>0.13</td></tr><tr><td>15.633</td><td>1015</td><td>0.02</td><td>62</td><td>0.03</td></tr><tr><td colspan="2">Totals</td><td>5947863</td><td>100.00</td><td>180143</td><td>100.00</td></tr></table>		Det166 Results					Time	Area	Area %	Height	Height %	8.033	68097	1.14	1413	0.78	9.950	5707101	95.95	173377	96.24	11.917	35431	0.60	1744	0.97	12.383	97022	1.63	2645	1.47	13.333	31081	0.52	668	0.37	14.567	8116	0.14	234	0.13	15.633	1015	0.02	62	0.03	Totals		5947863	100.00	180143	100.00
		Det166 Results																																																				
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15.633	1015	0.02	62	0.03																																																		
Totals		5947863	100.00	180143	100.00																																																	
SE-HPLC	Purity [% as IgG monomer]	≥ 90%	96%																																																			
	Aggregates [as % of total peak area]	≤ 6%	1%																																																			
	Other Impurities [%]	No individual impurity ≥ 3%	No individual impurity ≥ 3%																																																			

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		Certificate of Analysis Form			
TITLE Certificate of Analysis for Monoclonal Antibody Produced by Lpath					LOT NUMBER LP121416
DOCUMENT NUMBER FM-COA-13	REVISION 0.0.1	SUPERCEDED REVISION 0.0.0	ISSUE DATE OCT 31, 2013	EFFECTIVE DATE Nov 1, 2013	PAGE 4 of 5

Attribute	Test		Specifications	Results
	SDS-PAGE reduced	Purity [%]	≥ 90%	≥ 90%
		50 kDa Band	~65%	~67%
		25 kDa Band	~35%	~33%
		Other Impurities [%]	No individual impurity ≥ 3%	No individual impurity ≥ 3%
	Endotoxin		≤ 10.0 EU/mg of protein	0.06 EU/mg
	Bioburden		≤ 10 CFU/mL	Not done

Comments																																
Source Data: <table border="1" data-bbox="573 1117 1073 1377"> <thead> <tr> <th></th> <th>Lpath notebook</th> <th>Pages</th> </tr> </thead> <tbody> <tr> <td>Production</td> <td>277</td> <td>149</td> </tr> <tr> <td>Purification</td> <td>296</td> <td>11,18,25</td> </tr> <tr> <td>UV A280</td> <td>296</td> <td>48</td> </tr> <tr> <td>SDS-PAGE</td> <td>296</td> <td>26</td> </tr> <tr> <td>LPA binding ELISA</td> <td>221</td> <td>21-22</td> </tr> <tr> <td>IL-6</td> <td>237</td> <td>48-49</td> </tr> <tr> <td>EDG2</td> <td>237</td> <td>51</td> </tr> <tr> <td>SE-HPLC</td> <td>296</td> <td>33-34</td> </tr> <tr> <td>Endotoxin</td> <td>298</td> <td>13</td> </tr> </tbody> </table>				Lpath notebook	Pages	Production	277	149	Purification	296	11,18,25	UV A280	296	48	SDS-PAGE	296	26	LPA binding ELISA	221	21-22	IL-6	237	48-49	EDG2	237	51	SE-HPLC	296	33-34	Endotoxin	298	13
	Lpath notebook	Pages																														
Production	277	149																														
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SE-HPLC	296	33-34																														
Endotoxin	298	13																														
Additional Comments on Sample Quality:																																

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TITLE	Certificate of Analysis for Monoclonal Antibody Produced by Lpath				LOT NUMBER
					LP121416
DOCUMENT NUMBER	REVISION	SUPERCEDED REVISION	ISSUE DATE	EFFECTIVE DATE	PAGE
FM-COA-13	0.0.1	0.0.0	OCT 31, 2013	Nov 1, 2013	5 of 5

JUDGMENT

Approved for testing in humans

□ - Accepted

□ - Rejected

**For Research Only
Not For Use in Humans**

Prepared by: 
Cindy Dickerson, Ph.D.

Date: 12-6 2013

Reviewed by: Jon Wojciak, Ph.D.

Date: 12/6/ 2013

Approved by: Gary Woodnutt
Gary Woodnutt, Ph.D.

Date: 3rd Dec 2013

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ELISA Validation Report

Determination of LT3114 in Rat K₂EDTA Plasma By ELISA

ADDENDUM #1

QPS Project Number: 344-1401

Client Project Number: NA

Sponsor

Lpath Inc.

4025 Sorrento Valley Blvd.

San Diego, CA 92121

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QPS, LLC
Delaware Technology Park
3 Innovation Way, Suite 240
Newark, DE 19711
WEB: www.qps-usa.com
TEL: (302) 369-5601
FAX: (302) 369-5602

ELISA VALIDATION REPORT ADDENDUM #1

REPORT TITLE:

Determination of LT3114 in Rat K₂EDTA Plasma by ELISA

QPS PROJECT NUMBER:

344-1401

ORIGINAL REPORT DATE:

4 Sept 2014

ADDENDUM 1 DATE:

TBD

CONTRIBUTING SCIENTISTS:

Melissa Calvarese, B.S., Chad Asher, B.S.

TESTING FACILITY:

QPS, LLC
Delaware Technology Park
1 Innovation Way, Suite 200
Newark, DE 19711

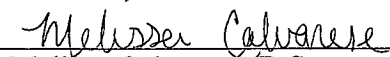
SPONSOR:

Lpath, Inc.
4025 Sorrento Valley Blvd.
San Diego, CA 92121

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1. SIGNATURE PAGE

WRITTEN BY:

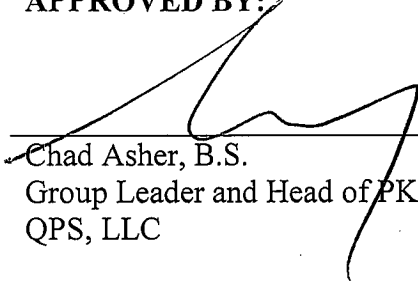


Melissa Calvarese, B.S.
Principal Investigator
QPS, LLC

10/27/14

Date

APPROVED BY:



Chad Asher, B.S.
Group Leader and Head of PK Bioanalysis
QPS, LLC

10/27/14

Date

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3. QUALITY ASSURANCE STATEMENT	6
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Long-Term Stability (LTS).....	7
5. TABLES.....	8
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Table 1B -70°C Long Term Frozen Stability for LT3114 in Rat K ₂ EDTA Plasma	10

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2. COMPLIANCE STATEMENT

This portion of the study was conducted in compliance with the Food and Drug Administration (FDA) Good Laboratory Practice Regulations (GLP) as set forth in Title 21 of the U.S. Code of Federal Regulations Part 58; and the FDA *Guidance for Industry: Bioanalytical Method Validation*, May 2001

To the best of my knowledge this report accurately reflects the raw data. I have reviewed the study and agree that the data supports the conclusions stated herein.

Melissa Calvarèse

Melissa Calvarèse, B.S.

Principal Investigator

QPS, LLC

10/27/14
Date

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3. QUALITY ASSURANCE STATEMENT

QPS 344-1401

Protocol Title: Determination of LT3114 in Rat K₂EDTA Plasma by ELISA

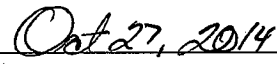
This QPS study has been audited and found to accurately reflect the requirements set forth by the 344-1401 validation protocol and standard operating procedures, and the results reported accurately reflect the source data. The following table summarizes the inspections and audits performed throughout the course of the study and the dates when each audit report and any findings were reported by QPS Quality Assurance.

Audit Description	QPS Auditor	Audit Dates	Date Audit Findings Reported to QPS Principal Investigator and Management
Study Records/ Draft Report Addendum 1	J. McClary	15 Oct 2014	15 Oct 2014

The undersigned has assured that the QPS final report accurately reflects the source data.



Jane McClary, B.A.
Quality Assurance
QPS, LLC



Date

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4. RESULTS AND DISCUSSION

Long-Term Stability (LTS)

LT3114 quality control samples prepared in Rat K₂EDTA Plasma were stored at -20°C and -70°C for 92 days before being analyzed against a freshly prepared calibration standard curve. The results (Table 1A and 1B) indicate LT3114 is stable for at least 92 days in Rat K₂EDTA Plasma at -20°C and -70°C.

5. TABLES

Table 1A**-20°C Long Term Frozen Stability for LT3114 in Rat K₂EDTA Plasma**

	Nominal Concentration (ng/mL)	
	300 Lot: MAC140708-12	1500 Lot: MAC140708-14
92 Days	276	1640
	268	1340
	277	1210
N	3	3
Mean	274	1400
SD	4.93	221
%CV	1.8	15.8
%RE	-8.7	-6.7

Initial run of QCs: 140709-3441401-RTPL-EEQ-1A-001,
to demonstrate proper preparation of LTS samples.

LTS Run ID: 141008-3441401-RTPL-MAC-1A-011

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Table 1B**-70°C Long Term Frozen Stability for LT3114 in Rat K₂EDTA Plasma**

	Nominal Concentration (ng/mL)	
	300 Lot: MAC140708-12	1500 Lot: MAC140708-14
92 Days	276	1440
	289	1520
	263	1370
N	3	3
Mean	276	1440
SD	13.0	75.1
%CV	4.7	5.2
%RE	-8.0	-4.0

Initial run of QCs: 140709-3441401-RTPL-EEQ-1A-001, to demonstrate proper preparation of LTS samples.

LTS Run ID: 141008-3441401-RTPL-MAC-1A-011

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